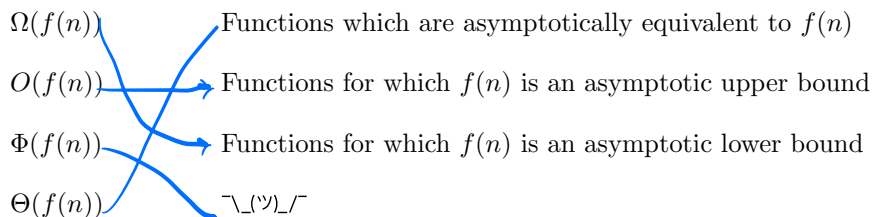


All students should watch the videos and review the readings accompanying this LGA in preparation for this assignment and the next lecture.

Individual self-test questions

All students should answer this set of questions. You will compare your answers with your group during the next lecture period. Note, we may from time to time include trick questions.

1. Match the symbols on the left with the descriptions on the right:



$$f(n) \leq c \cdot g(n)$$

Some of the next set of questions ask about rolls on dice. While 6-sided dice are traditional, some games use dice with different face counts, such as 4, 8, 10, 12, and 20. Assume that an n -sided die has faces numbered 1 - n .

2. The probability of seeing a 3 on one roll of a 6-sided die is $\frac{1}{6}$.

3. The probability of seeing an even number rolled on an 8-sided die is $\frac{1}{2}$.

4. The probability of rolling one 6-sided die and getting a 3, followed by rolling a second 6-sided die and getting a 2 is $\frac{1}{36}$.

$$\frac{1}{6} \cdot \frac{1}{6}$$

5. The probability of rolling a 3 and 2 on two 6-sided dice is $\frac{2}{36}$.

$$\begin{matrix} 3 & 2 \\ 2 & 3 \end{matrix} \Rightarrow \left(\frac{1}{6} \cdot \frac{1}{6} \right) + \left(\frac{1}{6} \cdot \frac{1}{6} \right)$$

6. A probability mass function is a function that assigns a real number to each possible outcome in an experiment.

random variable

Group questions

Distribute the following questions across the members of your group. You will share your solutions (and most importantly the *method* of your solutions) during the next lecture period. Divide up the questions so that each question has at least three solutions from different group members.

1. (Basic) In how many ways can a president, vice-president, and treasurer be selected from a club with 8 members if no one can hold more than one office?

P VP T

$$\frac{8!}{(8-3)!}$$

"order matters"

$$\frac{n!}{(n-k)!}$$

Jim Tom Alex
P VP T

$$= 336$$

2. (Basic) How many 5-digit binary numbers contain exactly 2 1's?

"five choose two"

$$\binom{5}{2} = \frac{5!}{2!(5-2!)} = \frac{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{2 \cdot 1 \cdot 3 \cdot 2 \cdot 1} = 10$$

3. (Intermediate) Suppose you roll a 6-sided die twice. What is the probability that you rolled a 3 at least once?

1 - no 3's

At 1 "3" = 1 - no 3's

$$1 - \left(\frac{5}{6} \cdot \frac{5}{6} \right) = 1 - \frac{25}{36} = \frac{11}{36}$$

4. (Basic) Assume I have in my pocket a nickel, a dime, 2 quarters, and a half-dollar. I pull a coin at random from my pocket. If we assign to the outcome the value of the coin I pull out, what is the expected value?

1 → 5 ¢

2 → 25 ¢

1 → 50 ¢

Avg. expected value

$$= 1 \cdot 0.05 + 2 \cdot 0.25 + 1 \cdot .5$$

The next set of questions concern the function COMPUTE:

function COMPUTE(n, m) // n and m are positive integers

$r \leftarrow 0$

for $i = 1$ **to** n **do** n

$x \leftarrow$ a number selected uniformly at random between 1 and m

$y \leftarrow 0$

for $j = 1$ **to** x **do** $[1, m]$

$y \leftarrow y + 1$

$r \leftarrow r \times m + y$

return r

* average outcome if you repeated pulling out coins

outer loop: n times

inner: $1 \leq x \leq m, x = \text{rand}(1, m)$

5. (Basic) What is the "big-O" complexity of COMPUTE in the **worst case**? Note this should be a function of both n and m . Briefly justify your answer.

$$O(n \cdot m) \text{ where } m = n.$$

6. (Basic) What is the "big-O" complexity of COMPUTE in the **best case**? Briefly justify your answer.

$$O\left(\left(\frac{1}{m}\right)^n\right) O(n) \text{ where } x=1 \text{ for all } n \text{ iterations.}$$

7. (Intermediate) What is the "big-O" complexity of COMPUTE in the **average case**? Briefly justify your answer.

$$O\left(n \cdot \left(\frac{m+1}{2}\right)\right)$$

* with enough randomness, the average x value should be

$$\therefore O(n \cdot m)$$

expected value: $\left(\frac{m+1}{2}\right)$

$g(n) \in O(f(n)) \rightarrow f(n)$ is the upper bound

8. (Basic) Formally show that $3n - 10 \in O(n)$. (Provide suitable choices for c and n_0 , then use any means you wish to prove the inequality: induction, a graphical plot, or some other convincing argument.)

End-goal:

$$3n - 10 \leq c \cdot n$$

for all $n \geq n_0$

$$3n - 10 \leq cn$$

$$3n - cn \leq 10$$

$$n(3 - c) \leq 10$$

$$n \cdot (3 - 4) \leq 10$$

$$n \cdot -1 \leq 10$$

$$\text{Let } c = 4 \text{ \& } n \geq 10.$$

9. (Intermediate) Formally show that $3n - 10 \in \Omega(n)$. (Provide suitable choices for c and n_0 , then use any means you wish to prove the inequality: induction, a graphical plot, or some other convincing argument.)

$$n(3 - c) \geq 10$$

$$n(3 - 2) \geq 10$$

$$n \cdot 1 \geq 10$$

$$\text{Let } c = 2 \text{ \& } n_0 = 10$$

for all $n \geq n_0$.

definitions:

• $g(n)$ is $\Omega(f(n))$

$$g(n) \geq c \cdot f(n)$$

• $g(n)$ is $O(f(n))$

$$g(n) \leq c \cdot f(n)$$

• $g(n)$ is $\Theta(f(n))$

$$c_1 \cdot f(n) \leq g(n) \leq c_2 \cdot f(n)$$

COS

LBJ

local

remote

- federated → "identical local repos"
 ↳ bi-directional syncing across all repos

- replication
 ↳ full clone from
 local to repo

- curl to s3 or f perms

- virtual repository
 ↳ single place to access collection

of other repos of identical
package

→ large file storage

↳ no versioning on binaries, only
for

• edge node

↳ read only environment



1	2	3	1
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$$\frac{1}{99} + \frac{1}{99}$$

1) $\frac{2}{99}$ because 2 select's

2)

$$\frac{1}{99} + \frac{1}{98} + \frac{1}{97} + \dots + \frac{1}{1}$$

$$\left(\frac{1}{99} + \frac{1}{99-n} \right)$$